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THREE ELECTRODE CELL INCLUDING SOLID ELECTROLYTE

Abstract

A three electrode cell includes an anode, a cathode, a solid electrolyte between the anode and the cathode, an elongated reference electrode embedded in the solid electrolyte, and a housing surrounding the anode, the cathode and the solid electrolyte. The reference electrode extends out of the housing.

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates generally to three electrode cells, and more specifically to

three electrode cells including solid electrolyte.

BACKGROUND

[0002] The performance of All Solid State Batteries (ASSB) are highly dependent on the quality of the anode and cathode laminate. However, in a regular full cell study, the performance of the anode and cathode are all combined with each other. Hence, it is very difficult to do the diagnostics of the cell based on electrochemical data to distinguish the issues from cathode and anode, especially for ASSB. Thus, three electrode cells are used in laboratory setting and include a reference electrode to test electrochemical data of the anode and cathode.

SUMMARY

[0003] A three electrode cell includes an anode, a cathode, a solid electrolyte between the anode and the cathode, an elongated reference electrode embedded in the solid electrolyte, and a housing surrounding the anode, the cathode and the solid electrolyte. The reference electrode extends out of the housing.

[0004] In examples, the reference electrode is a wire.

[0005] In examples, the wire has a diameter of between 0.1 and 0.5 mm.

[0006] In examples, the reference electrode is formed of lithiated aluminum.

[0007] In examples, the reference electrode passes entirely through the solid electrolyte and the housing with a first end of the reference electrode being outside of the housing and a second end of the reference electrode being outside of the housing.

[0008] In examples, the housing includes a hollow body including a bore receiving the anode, the cathode, and the solid electrolyte, the reference electrode extending through a first hole in the hollow body into the bore.

[0009] In examples, the reference electrode extends through a second hole in the hollow body into the bore such that a first end of the reference electrode and a second end of the reference electrode, opposite of the first end, are positioned outside of the hollow body.

[0010] In examples, the hollow body is formed of a ceramic material.

[0011] In examples, the housing further includes a first plunger configured for being inserted into a first end of the bore to contact the anode and a second plunger configured for being inserted into a second end of the bore to contact the cathode.

[0012] In examples, the first plunger and the second plunger are formed of stainless steel, tungsten, or nickel coated stainless steel.

[0013] In examples, the hollow body is formed of a ceramic material or PEEK.

[0014] In examples, the housing further includes a casing surrounding the hollow body, the casing including a first hole aligned with the first hole of the hollow body and a second hole aligned with the second hole of the hollow body, the reference electrode extending through the first hole and the second hole of the casing.

[0015] In examples, the casing includes a first section and a second section that are removably connected to each other, the housing further including a first plunger configured for being inserted through the first section of the casing and into a first end of the bore to contact the anode and a second plunger configured for being inserted through the second section of the casing and into a second end of the bore to contact the cathode, a first plunger seal being sandwiched between the first plunger and the first section of the casing, a second plunger seal being sandwiched between the second plunger and the second section of the casing.

[0016] In examples, the housing further includes a first adapter configured for being inserted through the casing and into the hollow body and a second adapter configured for being inserted through the casing and into the hollow body, the first adapter including a first hole configured for the reference electrode to pass through, the second adapter including a second hole configured for the reference electrode to pass through,

[0017] In examples, the first adapter includes a first inner adapter section configured for being inserted through the casing and into the hollow body and a second inner adapter section configured

for being inserted through the casing and into the hollow body, a first inner seal being sandwiched between the first inner adapter section and the hollow body, and a second inner seal being sandwiched between the second inner adapter section and the hollow body.

[0018] In examples, the housing further includes a first outer adapter section configured for being inserted into the first inner adapter section and a second outer adapter section configured for being inserted into the second inner adapter section, the first inner adapter section including a first hole forming a portion of the first hole of the first adapter, and the first outer adapter section including a first hole forming a portion of the first hole of the first adapter, the second inner adapter section including a second hole forming a portion of the second hole of the second adapter, and the second outer adapter section including a second hole forming a portion of the second hole of the second adapter, a first outer seal being sandwiched between the first inner adapter section and the first outer adapter section, and a second outer seal being sandwiched between the second inner adapter section and the second outer adapter section.

[0019] A method is also provided including stacking the anode, the cathode, and the solid electrolyte in a hollow body of the housing; and then inserting the reference electrode through the hollow body and the solid electrolyte.

[0020] In examples, the method further includes, after the stacking and before the inserting: installing a casing of the housing onto the hollow body; and inserting a first adapter through the casing and into contact with the hollow body and inserting a second adapter through the casing and into contact with the hollow body, the inserting of the reference electrode including inserting a first end of the reference electrode through the first adapter, through the casing, through the hollow body and through the second adapter.

[0021] In examples, the method further includes, prior to the stacking, installing a portion of a first plunger into the hollowing body, the stacking including stacking the anode, the cathode, and the solid electrolyte in the hollow body on the portion of the first plunger.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0022] The present disclosure is described below by reference to the following drawings, in which:

[0023] FIG. 1 shows a radial cross-sectional view of a three electrode cell according to the present disclosure;

[0024] FIG. 2 schematically shows an anode, a cathode and a solid electrolyte of the three electrode cell; and

[0025] FIG. 3 shows an exploded view of the three electrode cell.

DETAILED DESCRIPTION

[0026] FIG. 1 shows a radial cross-sectional view of a three electrode cell **10** according to the present disclosure; FIG. 2 schematically shows an anode **12**, a cathode **14** and a solid electrolyte **16** of three electrode cell **10**, and FIG. 3 shows an exploded view of three electrode cell **10**. An elongated reference electrode **18**, which may be a wire, is embedded in the solid electrolyte **16**. Solid electrolyte can be added on both sides to compress the reference electrode **18**. A housing **20** surrounds the anode **12**, the cathode **14**, and the solid electrolyte **16**. The reference electrode **18** extends out of the housing **20**. More specifically, the reference electrode **18** passes entirely through the solid electrolyte **16** and the housing **20** with a first end **18a** of the reference electrode **18** being outside of the housing **20** and a second end **18b** of the reference electrode **18** being outside of the housing **20**.

[0027] In some examples, the reference electrode **18** is formed of lithiated aluminum, with an entire outer surface of the electrode being lithiated. The reference wire can have a diameter of between 0.1 and 0.5 mm.

[0028] The housing **20** may include a hollow body **22** comprising a bore **22a** for receiving the anode **12**, the cathode **14**, and the solid electrolyte **16**, with the reference electrode **18** extending through a first hole **22b** and a second hole **22c** in the hollow body **22**. The hollow body **22** may be a ceramic material or PEEK.

[0029] The housing **20** may further include a first plunger **24** configured for being inserted into one end of the bore **22a** to contact the anode **12**, and a second plunger **26** configured for being inserted into the opposite end of the bore **22a** to contact the cathode **14**. The plungers **24**, **26** can be made of stainless steel, tungsten, or nickel coated stainless steel.

[0030] The housing **20** may further include a casing **28** surrounding the hollow body **22**. The casing **28** can include a first section **30** and a second section **32**, which are removably connected to each other by threads. Sections **30**, **32** each have a cup shape and first section **30** is screwed into second section **32** by the outer threads of first section **30** intermeshing with the inner threads of second section **32**. Plungers **24**, **26** extend through respective holes **30a**, **30b** in casing **28** such that an end of each of plungers **24**, **26** contacts the anode **12** and cathode **14**, respectively. To seal the interfaces between plungers **24**, **26** and casing **28**, plunger seals **34**, **36** are provided at these interfaces.

[0031] More specifically, first section **30** includes the hole **30a** and second section **32** includes the hole **32a**, such that plunger **24** extends through first section **30** and plunger **26** extends through second section **32**. Plungers **24**, **26** each include a pin **24a**, **26a** forming a narrower portion thereof and a plate **24b**, **26b** forming a wider portion thereof. Pins **24a**, **26b** extend through the respective hole **30a**, **32a** into bore **22a** and contact the anode **12** and cathode **14**, respectively. Plunger seals **34**, **36** contact the outer surface of the respective pin **24a**, **24b** and contact casing **28** adjacent to the respective holes **30a**, **32a**. Plates **24b**, **26b** are positioned outside of casing **28**.

[0032] The housing **20** further includes adapters **38**, **40** for insertion through the casing **28** into the hollow body **22**. The housing **20** includes aligned holes allowing the reference electrode **18** to extend through the adapters **38**, **40**, the casing **28** and the hollow body **22**. More specifically, Casing **28** includes holes **28a**, **28b**, each receiving a respective one of the adapters **38**, **40**, hollow body **22** includes first hole **22b** and second hole **22c** through which reference electrode **18** passes in the assembled orientation of three electrode cell **10** and adapters **38**, **40** each include a respective hole **38a**, **40a** through which reference electrode **18** passes in the assembled orientation of three electrode cell **10**.

[0033] First adapter **38** includes a first inner adapter section **42** and a first outer adapter section **46**, along with a first inner seal **50** and a first outer seal **54**, while second adapter **40** includes a second inner adapter section **44** and a second outer adapter section **48**, along with a second inner seal **52** and a second outer seal **56**.

[0034] First inner adapter section **42** and second inner adapter section **44** are inserted into opposite sides of casing **28** and pass through casing **28** into contact with opposite sides of hollow body **22**. Hollow body **22** includes recesses **22d**, **22e** on opposite sides of hollow body **22**, each for receiving an end of a respective one of first inner adapter section **42** and second inner adapter section **44** such that the end of the respective section **42**, **44** contacts a surface **22f**, **22g** surrounding the respective hole **22b**, **22c**. Inner seals **50**, **52** are held in recesses **22d**, **22e** and abut the ends of inner adapter sections **42**, **44**. Inner adapter sections **42**, **44** are removably connected to casing **28** in holes **28a**, **28b**. Specifically, inner adapter sections **42**, **44** are removably connected to casing **28** by threads on the outer circumference of inner adapter sections **42**, **44** engaging threads on the inner circumferential of holes **28a**, **28b**.

[0035] First outer adapter section **46** and second outer adapter section **48** are inserted into ends of first inner adapter section **42** and **44**, respectively, that face away from hollow body **22** in the assembled orientation of three electrode cell **10**. More specifically, each of first inner adapter section **42** and second inner adapter section **44** includes a respective recess **42a**, **44a**, in the end facing away from the end that contacts hollow body **22**, into which the respective outer adapter

section **46**, **48** is inserted.

[0036] A method for constructing the three electrode cell **10** may include stacking the anode **12**, cathode **14**, and solid electrolyte **16** within the hollow body **22** of the housing **20** and inserting the reference electrode **18** through the solid electrolyte **16**. The method can include installing the casing **28** onto the hollow body **22** and inserting the adapters **38**, **40** into the casing **28** and in contact with the hollow body **22**. Prior to the stacking of the anode **12**, cathode **14**, and solid electrolyte **16** within the hollow body **22**, the method can include installing a portion—i.e., pin **24a**—of the plunger **24** into the hollow body **22** and resting plate **24b** of plunger on a support surface **58**. After the anode **12**, cathode **14**, and solid electrolyte **16** are stacked within the hollow body **22**, the pin **26a** of plunger **26** is inserted into hollow body. The inserting of the reference electrode **18** includes inserting the first end **18a** of the reference electrode **18** through the first adapter **38**, through the casing **28**, through the hollow body **22** and through the second adapter **40**. After this insertion, the first end **18a** protrudes in a first direction from the adapter **40** and the second end **18b** protrudes in a second direction, opposite of the first direction, from the adapter **38** such that both ends **18a**, **18b** are outside of the housing **20**. Protruding on both side secures the position of the reference electrode, which greatly helps the accuracy of the test.

[0037] The electron flow initiated by a test using three electrode cell **10** is continuous flow from anode **12** to cathode **14** during charging and from cathode **14** to anode **12** during discharging. Reference electrode **18** is used to detect the potential difference. When the test is started, current is applied between anode **12** to cathode **14**, and the reference electrode **18** is used to monitor the voltage change of anode **12** to cathode **14** at the same time.

[0038] In the preceding specification, the disclosure has been described with reference to specific exemplary embodiments and examples thereof. It will, however, be evident that various modifications and changes may be made thereto without departing from the broader spirit and scope of disclosure as set forth in the claims that follow. The specification and drawings are accordingly to be regarded in an illustrative manner rather than a restrictive sense.

Reference Numerals

[0039] **10** three electrode cell [0040] **12** anode [0041] **14** cathode [0042] **16** solid electrolyte [0043] **18** reference electrode [0044] **18a** first end [0045] **18b** second end [0046] **20** housing [0047] **22** hollow body [0048] **22a** bore [0049] **22b** first hole [0050] **22c** second hole [0051] **22d** recesses [0052] **22e** recesses [0053] **22f** surface [0054] **22g** surface [0055] **24** plungers [0056] **24a** pin [0057] **24b** plate [0058] **26** plungers [0059] **26a** pin [0060] **26b** plate [0061] **28** casing [0062] **28a** holes [0063] **28b** holes [0064] **30** first section [0065] **30a** respective holes [0066] **30b** respective holes [0067] **32** second section [0068] **32a** hole [0069] **34** plunger seals [0070] **36** plunger seals [0071] **38** adapters [0072] **38a** respective hole [0073] **40** adapters [0074] **40a** respective hole [0075] **42** first inner adapter section [0076] **42a** respective recess [0077] **44** second inner adapter section [0078] **44a** respective recess [0079] **46** first outer adapter section [0080] **48** second outer adapter section [0081] **50** first inner seal [0082] **52** second inner seal [0083] **54** first outer seal [0084] **56** second outer seal [0085] **58** support surface

Claims

1. A three electrode cell including: an anode; a cathode; a solid electrolyte between the anode and the cathode; an elongated reference electrode embedded in the solid electrolyte; and a housing surrounding the anode, the cathode and the solid electrolyte, the reference electrode extending out of the housing.
2. The three electrode cell as recited in claim 1 wherein the reference electrode is a wire.
3. The three electrode cell as recited in claim 2 wherein the wire has a diameter of between 0.1 and 0.5 mm.
4. The three electrode cell as recited in claim 1 wherein the reference electrode is formed of

lithiated aluminum.

5. The three electrode cell as recited in claim 1 wherein the reference electrode passes entirely through the solid electrolyte and the housing with a first end of the reference electrode being outside of the housing and a second end of the reference electrode being outside of the housing.
6. The three electrode cell as recited in claim 1 wherein the housing includes a hollow body including a bore receiving the anode, the cathode, and the solid electrolyte, the reference electrode extending through a first hole in the hollow body into the bore.
7. The three electrode cell as recited in claim 6 wherein the reference electrode extends through a second hole in the hollow body into the bore such that a first end of the reference electrode and a second end of the reference electrode, opposite of the first end, are positioned outside of the hollow body.
8. The three electrode cell as recited in claim 6 wherein the hollow body is formed of a ceramic material.
9. The three electrode cell as recited in claim 6 wherein the housing further includes a first plunger configured for being inserted into a first end of the bore to contact the anode and a second plunger configured for being inserted into a second end of the bore to contact the cathode.
10. The three electrode cell as recited in claim 9 wherein the first plunger and the second plunger are formed of stainless steel, tungsten, or nickel coated stainless steel.
11. The three electrode cell as recited in claim 6 wherein the hollow body is formed of a ceramic material or PEEK.
12. A three electrode cell including: an anode; a cathode; a solid electrolyte between the anode and the cathode; an elongated reference electrode embedded in the solid electrolyte; and a housing surrounding the anode, the cathode and the solid electrolyte, the reference electrode extending out of the housing, the housing including a hollow body including a bore receiving the anode, the cathode, and the solid electrolyte, the reference electrode extending through a first hole in the hollow body into the bore, the housing further includes a casing surrounding the hollow body, the casing including a first hole aligned with the first hole of the hollow body and a second hole aligned with the second hole of the hollow body, the reference electrode extending through the first hole and the second hole of the casing.
13. The three electrode cell as recited in claim 12 wherein the casing includes a first section and a second section that are removably connected to each other, the housing further including a first plunger configured for being inserted through the first section of the casing and into a first end of the bore to contact the anode and a second plunger configured for being inserted through the second section of the casing and into a second end of the bore to contact the cathode, a first plunger seal being sandwiched between the first plunger and the first section of the casing, a second plunger seal being sandwiched between the second plunger and the second section of the casing.
14. The three electrode cell as recited in claim 12 wherein the housing further includes a first adapter configured for being inserted through the casing and into the hollow body and a second adapter configured for being inserted through the casing and into the hollow body, the first adapter including a first hole configured for the reference electrode to pass through, the second adapter including a second hole configured for the reference electrode to pass through,
15. The three electrode cell as recited in claim 14 wherein the first adapter includes a first inner adapter section configured for being inserted through the casing and into the hollow body and a second inner adapter section configured for being inserted through the casing and into the hollow body, a first inner seal being sandwiched between the first inner adapter section and the hollow body, and a second inner seal being sandwiched between the second inner adapter section and the hollow body.
16. The three electrode cell as recited in claim 15 wherein the housing further includes a first outer adapter section configured for being inserted into the first inner adapter section and a second outer adapter section configured for being inserted into the second inner adapter section, the first inner

adapter section including a first hole forming a portion of the first hole of the first adapter, and the first outer adapter section including a first hole forming a portion of the first hole of the first adapter, the second inner adapter section including a second hole forming a portion of the second hole of the second adapter, and the second outer adapter section including a second hole forming a portion of the second hole of the second adapter, a first outer seal being sandwiched between the first inner adapter section and the first outer adapter section, and a second outer seal being sandwiched between the second inner adapter section and the second outer adapter section.

17. A method of constructing a three electrode cell comprising: stacking an anode, a cathode, and a solid electrolyte in a hollow body of a housing, the solid electrolyte being between the anode and the cathode; and inserting an elongated reference electrode through the hollow body and the solid electrolyte.

18. The method as recited in claim 17 further comprising, after the stacking and before the inserting: installing a casing of the housing onto the hollow body; and inserting a first adapter through the casing and into contact with the hollow body and inserting a second adapter through the casing and into contact with the hollow body, the inserting of the reference electrode including inserting a first end the reference electrode through the first adapter, through the casing, through the hollow body and through the second adapter.

19. The method as recited in claim 17 further comprising, prior to the stacking, installing a portion of a first plunger into the hollowing body, the stacking including stacking the anode, the cathode, and the solid electrolyte in the hollow body on the portion of the first plunger.
